

Deep Interest Evolution Network - DIEN

Saioa Prieto Portugal, 633301

Alexander Pérez Bunea, 636393

Fabio Misino, 512377

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Link to the official paper: [PAPER](#)

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INTRODUCTION

Click-Through Rate (CTR) prediction plays a pivotal role in online advertising, influencing both platform revenue and user satisfaction. Despite its importance, **many existing models fall short in capturing the complexity of user behaviour**. They often **treat user actions as static**, ignoring the dynamic nature of individual preferences. Additionally, these models frequently do **not separate explicit behaviours**—such as clicks or purchases—**from underlying latent interests**, which are more intricate and evolve over time. This lack of distinction restricts the model's ability to accurately predict future interactions. Furthermore, traditional approaches typically **fail to model the temporal evolution of user interests**, leading to suboptimal personalization and reduced advertising effectiveness. That is why the following architecture was investigated.

MODEL'S ARCHITECTURE

Models' main idea

To address the limitations of traditional CTR prediction models, the authors introduce a **novel architecture called DIEN (Deep Interest Evolution Network)**. Unlike previous approaches that treat user behaviour as static, DIEN is specifically **designed models interest evolution, not just static interest**. It first **learns latent user interests from behaviour sequences**, going beyond surface-level actions to understand the motivations behind them. Then, it **models how these interests evolve over time**, using sophisticated mechanisms to track changes and transitions in user preferences. By aligning the evolving interests with the target item, DIEN significantly **enhances the relevance and accuracy of CTR predictions**, leading to improved ad targeting and overall performance in online advertising systems.

DIEM Architecture

The DIEN architecture is built upon two main layers:

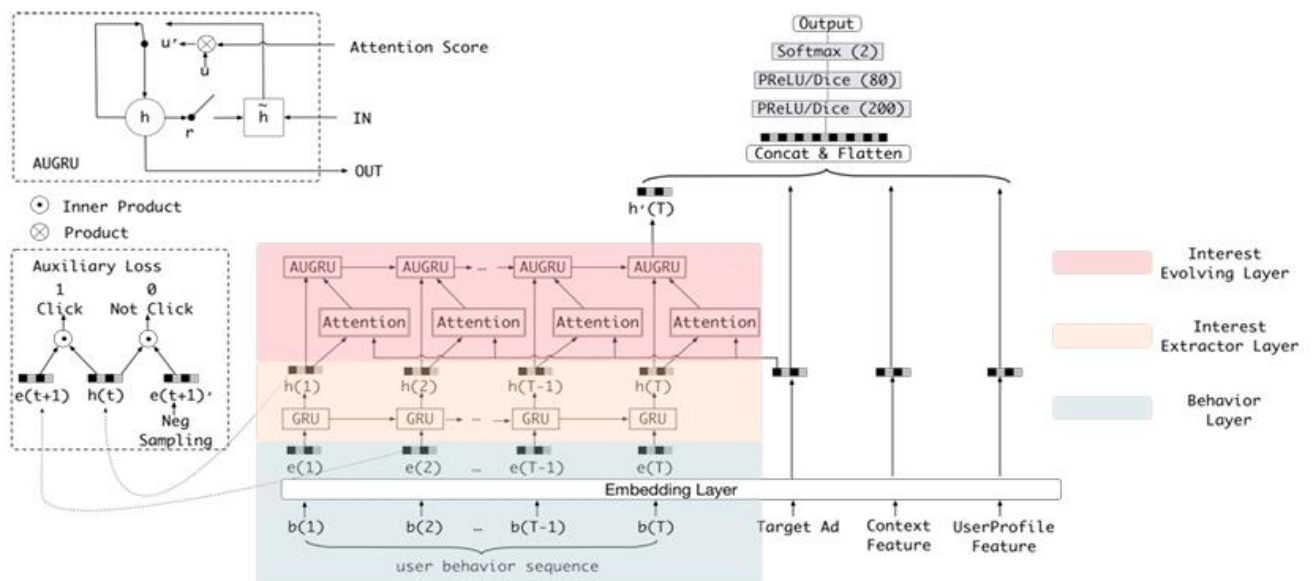
1. Interest Extractor Layer:

This layer utilizes a Gated Recurrent Unit (GRU) to process the user's historical behaviour sequence. To ensure that the model captures meaningful interest representations at every time step—not just the final output—an **auxiliary loss** is introduced. This auxiliary loss supervises each hidden state by predicting the user's next behaviour, effectively guiding the model to learn **transitions between latent interests**.

over time. The total loss function combines the standard CTR prediction loss with the auxiliary loss, encouraging stronger and more informative hidden state representations. As a result, each hidden state serves as a robust encoding of the user's latent interest at a specific point in the sequence.

2. Interest Evolving Layer:

Once latent interests are extracted, this layer models how they evolve **in relation to a specific target item**. DIEN introduces a novel mechanism called **AUGRU (Attentional Update Gate GRU)**, which enhances the traditional GRU by integrating an attention mechanism into the update gate. This attention is calculated with respect to the target item and modulates the update gate on a **dimension-wise basis**, allowing for fine-grained control over how interests are updated. By doing so, AUGRU selectively emphasizes relevant interests and filters out irrelevant ones, enabling the model to more accurately track the **target-specific evolution** of user preferences.



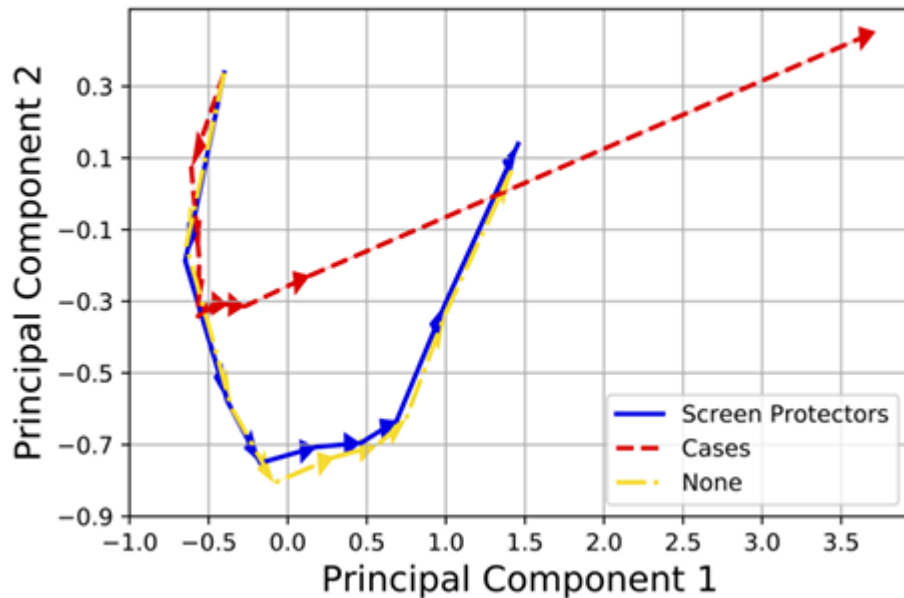
Data flow

The data flow of the DIEN (Deep Interest Evolution Network) model begins with a **sequence of user behaviours**, such as clicked items, which are sorted in chronological order. Each behaviour, initially represented as a one-hot vector $b[t]$, is transformed into a dense embedding vector $e[t]$ through the embedding layer. These embeddings are then **passed into the Interest Extractor Layer**, which uses a **GRU** to model the sequence and **generate hidden states** $h[t]$ that capture the user's latent interests at each time step. To **enhance the quality** of these hidden states, DIEN uses the **auxiliary loss** that

supervises each hidden state by **predicting the next behaviour** in the sequence. This encourages the model to learn meaningful and temporally-aware interest representations, not just final outputs. Next, **the sequence of hidden states enters the Interest Evolving Layer**, where the novel **AUGRU** processes them. AUGRU incorporates **attention scores based on the target item** into its update mechanism, allowing the model to emphasize relevant interests and filter out irrelevant ones. This layer **outputs a final interest state $h[T]$** that reflects the user's current intent in relation to the target item. Finally, this evolved **interest state is concatenated with the embedding vectors of remaining features** (such as user profile or target item embeddings), and **the combined representation is fed into a multi-layer perceptron (MLP)** for final click-through rate prediction.

Visualization

In the DIEN model, the authors visualize how user interests evolve over time by examining the hidden states produced by the AUGRU layer. These hidden states reflect the user's dynamic interest at each point in their behaviour history. To make this evolution interpretable, the high-dimensional hidden states are projected into a two-dimensional space using Principal Component Analysis (PCA) and then linked sequentially to show the trajectory of interest evolution. This visualization is performed for different target items to observe how the target affects the evolution path. When no specific target is used (i.e., attention scores are equal), the resulting path—the yellow curve—shows the natural progression of interest based solely on behaviour history. When the target item belongs to a category unrelated to the user's recent behaviour, such as "Screen Protectors," the evolution path (blue curve) closely resembles the yellow one, indicating minimal influence. However, when the target item is highly related to the user's recent activity—for example, if the last behaviour involved "Cases" and the target item also belongs to that category—the red curve shows a significant shift in the final hidden state. This indicates that the model assigns a high attention score to the most relevant behaviour, strongly influencing the interest representation. The visualization thus confirms that AUGRU effectively modulates interest evolution based on the relevance of the target item, allowing DIEN to focus on the most meaningful parts of user history for better CTR prediction.



Experiments done

The experiments conducted in the DIEN paper demonstrate the model's effectiveness compared to several state-of-the-art CTR prediction architectures. The authors evaluated DIEN on both public datasets (Amazon Books and Electronics) and a large-scale industrial dataset from Taobao's display advertising system. On public datasets, **DIEN significantly outperformed baseline models such as Wide & Deep, PNN, DIN, and GRU-based attention models.** Specifically, DIEN achieved AUC scores notably higher than all other competitors. These results affirm that DIEN's ability to model both latent interests and their temporal evolution leads to more accurate predictions, **Furthermore, exhaustive studies confirmed the individual benefits of the auxiliary loss and the novel AUGRU mechanism.**

CONCLUSION

This study shows that there are still many ways to improve click-through rate prediction by going beyond traditional models. DIEN demonstrates that focusing on how user interests evolve over time—rather than treating behavior as static—can lead to significantly better results. By introducing mechanisms like the auxiliary loss and AUGRU, the model captures user intent more accurately and adapts better to real-world dynamics. This suggests that deep learning models with more

personalized and time-sensitive approaches can greatly enhance the performance of recommendation and advertising systems.